MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 2J55

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Fixed frequency, 9,375 ± 30 MHz, pulse type, integral magnet, rated peak power output 40 kw.

ABSOLUTE RATINGS:

Parameter:	Ef	tk	epy	ib	Pi	pi	tpc	Du	T (anode)	Alt
Unit:	V	sec	kv	a	W	kw	μs	---	°C	ft
Maximum:	6.93	---	16	16	180	230	2.5	0.001	100	10,000
Minimum:	---	120	---	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS: See figure 1

Input connection: See figure 1

Output coupling: Mates with UG52B/U flange (MIL-F-3922/59)

Magnet isolation: See note 2

WEIGHT: 4 pounds (approximate)

MOUNTING POSITION: Any

TEST CONDITIONS: See note 2

Parameter:	Ef	tk	tpc	trv	Du	Ib	VSWR	T (anode)
Unit:	V	sec	μs	μs	---	mAdc	---	°C
Tolerance:	---	---	±10%	---	---	---	---	---
Test 1:	0	120(min)	1.0	0.2(max)	0.001	12	1.15(max)	80(approx)
Test 2:	2.5 (see note 3)	120(min)	2.0	0.2(max)	0.00065	8	1.15(max)	80(approx)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	---					
1301	Heater current	---	---	Ef = 6.3 V	If	0.90	1.10	A
4306	Pulse voltage	---	1		epy	11	13	kv
4003	Pressurizing	---	---	40 to 45 psia	---	---	---	---
4250	Power output	---	1	f = 300 (max)	Po	40	---	W
4308	RF bandwidth	---	1	Ib = 11 to 13 mAdc	BW	---	3 (pc	MHz
4310	Pulling factor	---	1	VSWR = 1.5	ΔF	---	15	MHz
4315	Stability	5	2	VSWR = 1.1 (max)	MP	---	1	%
---	Fixed frequency	---	1		F	9,345	9,405	MHz
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	---	---	No voltages	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	---	1	Group D	t	500	---	hrs
---	Life-test end points							
4250	Power output	---	1		Po	30	---	W
4308	RF bandwidth	---	1		BW	---	3	MHz
	Periodic-check tests							
1031	High-frequency vibration	---	---	No voltages			---	---
4027	Temperature coefficient	---	1		ΔF ΔT	---	0.25	MHz °C
1026	Low-temperature operation	---	1	tk = 120 (max) TA = -65°C	---	---	---	---
1002	Barometric pressure, reduced	---	1	Pressure = 380 mmHg (abs)	---	---	---	---
4315	Snap-on stability	5	2	VSWR = 1.1 (max)	MP	---	1	
1143	Forced cooling	4,u	1	TA = 25°, 5°C	T	---	50	°C

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NOTES:

1. Unless otherwise specified the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

2. In handling and mounting the tube, care shall be taken to prevent demagnetization. Magnetic materials or devices shall not be brought within 2 inches of the tube.

3. The product of the pulse duration in seconds and the pulse recurrence frequency in hertz shall not exceed 0.0007 for pi = 150 kw, or greater, and shall not exceed 0.001 in any circumstance. The pulse recurrance frequency shall be 325 pps, or greater. During high-voltage pulsed operation, the heater voltage should be varied with input power approximately in accordance with the formula:

$$Ef = 6.3\sqrt{1 - \frac{Pi}{150}}$$

--- with Ef = 0 for all values of Pi greater than 150. The heater should be protected against input pulse power by an appropriate filter.

4. With an airflow, at standard atmospheric pressure, of 65 cfm directed at the cooling fins from an orifice of 1-11/64 (29.77 mm) by 1-23/64 (34.53 mm) inches, the rise above the ambient temperature specified shall not be exceeded. The orifice shall be located 1/4 (6.35 mm) inch from the cooling fins.

5. The missing pulses shall be counted during any 3 minutes of a test interval of 6 minutes, or less. A missing pulse is defined as one whose energy within a ±1 percent frequency range, of the normal test frequency, is 70 percent or less than a normal pulse.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17.
DSA - ES

User activities:
Army - MU
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-24)

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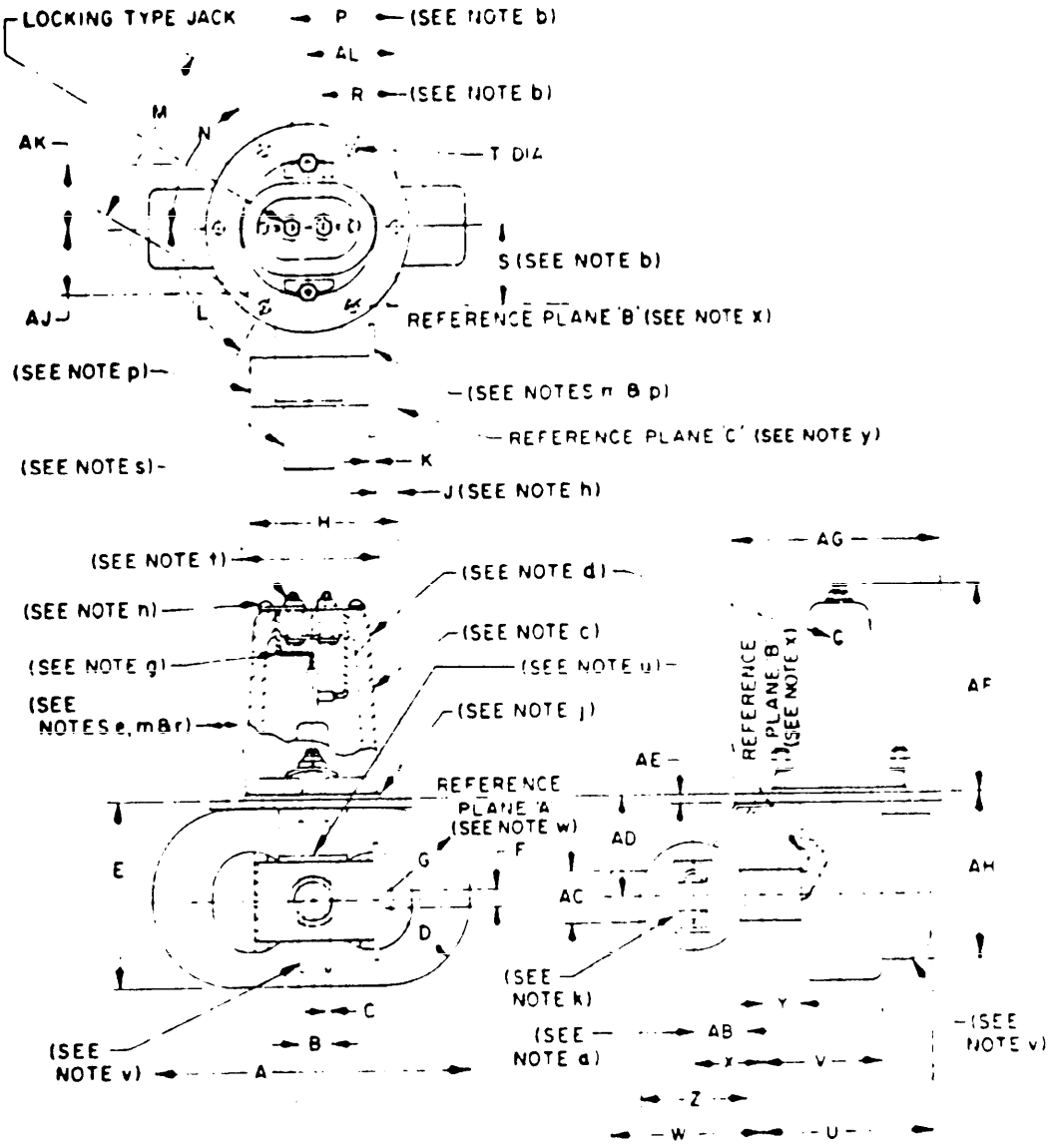


FIGURE 1. Outline drawing of electron tube type 2J55.

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NOTES:

a. Output flange to be cconcentric with the open end of the guide to within .010 (.25 mm).
b. The jack holes shall be within a radius of .023(.58 mm) of true location specified but shall be spaced .500(12.70 mm)±.010 (.25 mm) with respect to each other. The center lines of the holes shall be perpendicular to reference plane 'A' within 3°.
c. Pyrex glass, porcelain, or approved equivalent.
d. Indicate common cathode connection by letter C.
e. With the flange resting on a plane surface, the flatness of the mounting plate .500 (12.70 mm) from the outer edge shall be such that a .010(.25 mm) thickness gage .125(3.18 mm) wide shall not enter for a distance of more than .250(6.35 mm).
f. Metal surfaces shall be covered by painted finish except inside of waveguide, jack terminals and surfaces, covered by note m.
g. Filament leads shall be flexible and slack.
h. Tolerance includes angular as well as lateral deviations of the surface.
j. .031(.79 mm) asbestos gasket (optional).
k. Output to mate with UG52B/U flange (MIL-F-3922/59).
m. Shall be free from paint.
n. Leads soldered to terminal clip.
p. All joints in waveguide assembly shall be vacuum tight to provide hermetic seal at flange.
r. All joints in mounting plate shall be vacuum tight to provide hermetic seal.
s. Heater connection.
t. Any portion of the assembly extending above this surface shall be within a 1.109(28.17 mm) radius of the true center of the plate.
u. Anode temperature measurement point.
v. Radiator fins.
w. Reference plane A is defined as a plane passing along the surface of mounting plate as shown.
x. Reference plane B is defined as a plane passing through the center of the two mounting plate holes at plane A and is perpendicular to plane A.
y. Reference plane C is defined as a plane passing through the center of the mounting plate hole (as shown) at plane A and is mutually perpendicular to planes A and B.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note 1)		
J	.417 (10.59)	.457 (11.61)
Z	1.743 (44.27)	1.757 (44.63)
Quality conformance inspection, part 2		
A		5.375(136.53)
B		.500 (12.70)
E		3.125 (79.38)
H		2.500 (63.50)
K	.080 (2.03)	.090 (2.29)
L	59°48'	60°12'
M	2.869 (72.87)	2.881 (73.18)
N	59°48'	60°12'
T	.190 (4.83)	.196 (4.98)
U		2.745 (69.72)
W		2.661 (67.59)
X	1.172 (29.77)	1.212 (30.78)
Y	.432 (10.97)	
AB	.859 (21.82)	.891 (22.63)
AF	2.922 (74.22)	3.047 (77.39)
AG	3.219 (81.76)	3.281 (83.34)
AH		2.969 (75.41)
AJ	.875 (22.23)	.937 (23.80)
AK	.875 (22.23)	.937 (23.80)
AL	1.406 (35.71)	1.468 (37.29)
Quality conformance inspection, part 3 (periodic check)		
AD	1.542 (39.17)	1.582 (40.18)
AE	.120 (3.05)	.130 (3.30)
Reference dimensions		
C	.062 (1.57)	
D	1.344 (34.14)	
F	.374 (9.50)	
G	1.344 (34.14)	
P	1.687 (42.85)	
R	1.187 (30.15)	
S	1.245 (31.62)	
V	2.057 (52.25)	
AC	.750 (19.05)	

FIGURE 1. Outline drawing of electron tube type 2J55 - Continued.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-28264

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